

TG-WSPIS: Multi-Granular Text-Guided Weakly Supervised Pathology Image Segmentation

Text-Guided Weak Supervision for Computational Pathology

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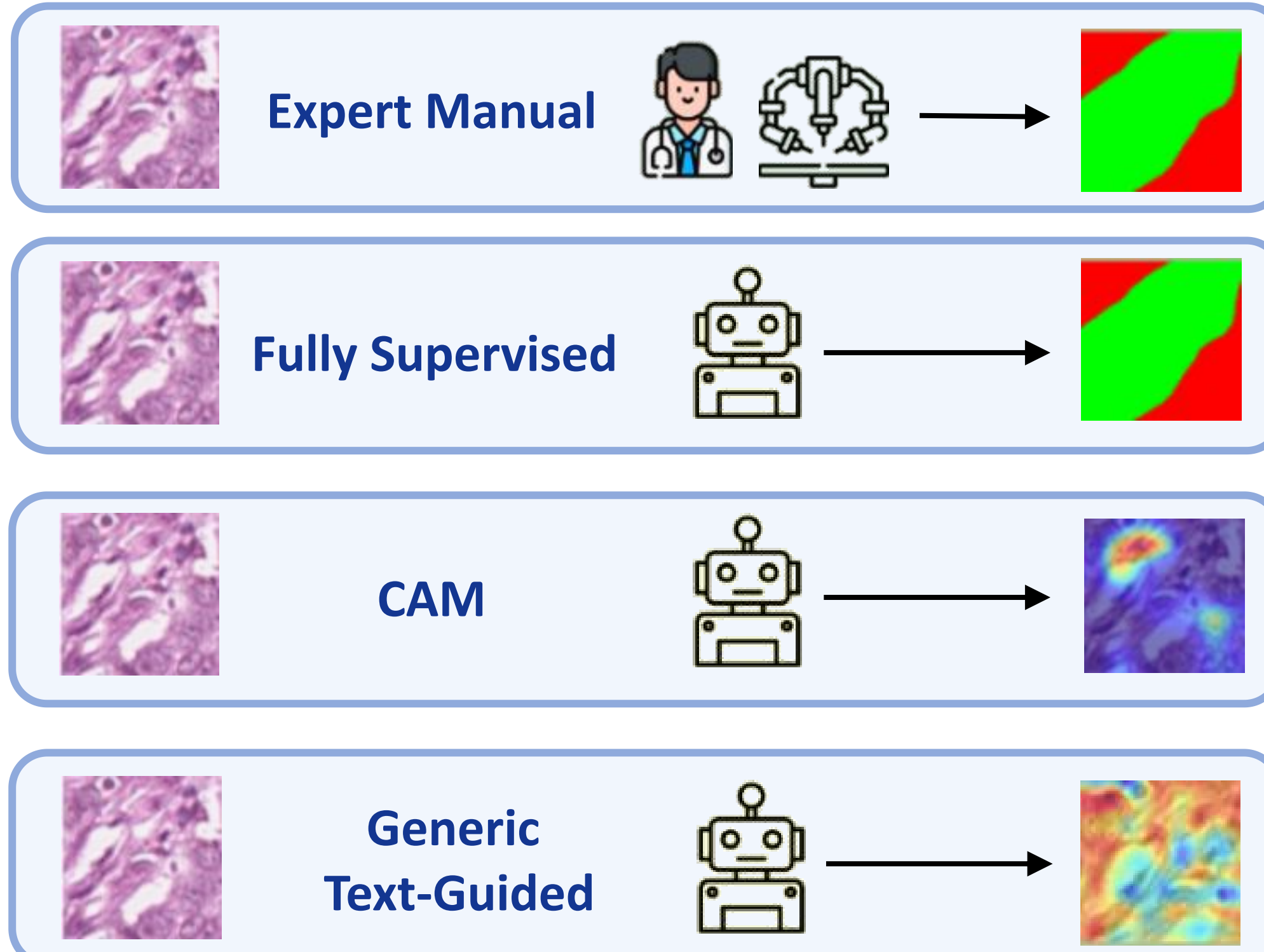
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1 Background / Motivation

- Pixel-level pathology annotation is costly, labor-intensive, and suffers from inter-expert inconsistency.
- CAM-based weak supervision only highlights discriminative regions, leaving scattered tissues under-segmented.
- Generic text-guided methods struggle with fine-grained spatial & quantitative cross-modal alignment.



2 Core Idea / Contributions



Multi-granular Text Guidance Paradigm

Image-specific captions at 3 granularities: global proportion, spatial distribution, and fused information.



TG-WSPIS Framework with PMCC & MI Learning

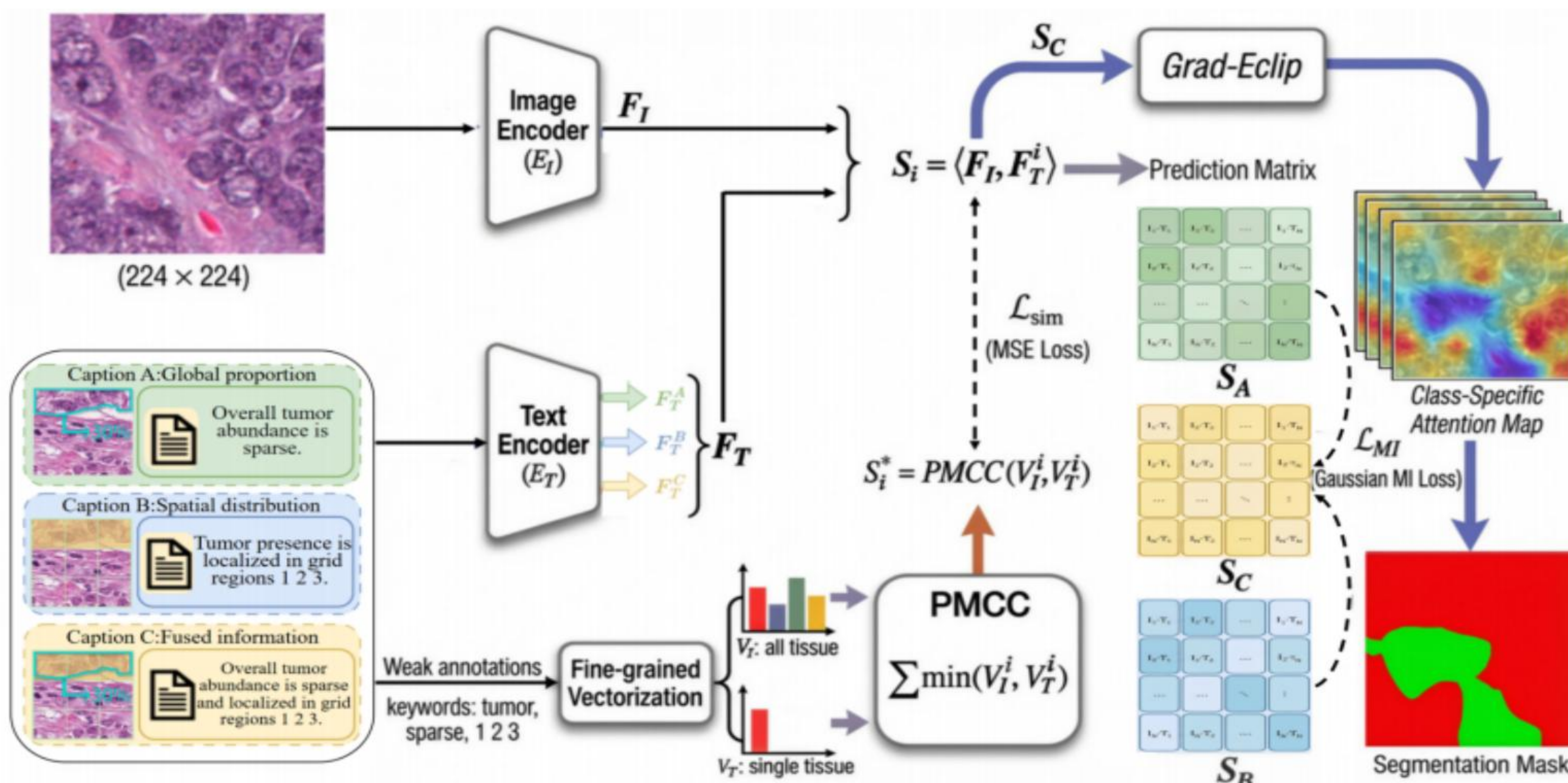
Soft quantifiable image-text alignment via PMCC, cross-granular semantic fusion via unidirectional Gaussian MI collaborative learning.



Two Standardized Benchmarks

BCSS-T3G and WSSS4LUAD-T3G for text-guided weakly supervised pathology segmentation.

3 Method Overview



4 Benchmarks / Datasets



BCSS-T3G:

TUM, STR, LYM, NEC



WSSS4LUAD-T3G:

TUM, STR, NOM



Both datasets are augmented with three-level captions for text-guided weak supervision.

5 Main Results

BCSS-T3G		WSSS4LUAD-T3G	
Method	mIoU(%)	Method	mIoU(%)
HistoSegNet	35.19	HistoSegNet	36.71
SEAM	63.38	SEAM	62.62
WSSS-Tissue	63.94	WSSS-Tissue	62.72
OEEM	64.35	OEEM	63.24
HAMIL	64.64	HAMIL	63.40
SAM3	64.83	SAM3	63.19
TPRO	68.04	TPRO	67.73
ARML	71.56	ARML	70.40
Ours(TG-WSPIS)	76.79	Ours(TG-WSPIS)	75.35

Outperforms SOTA ARML: **+5.23%** (BCSS-T3G), **+4.95%** (WSSS4LUAD-T3G)

6 Ablation / Analysis

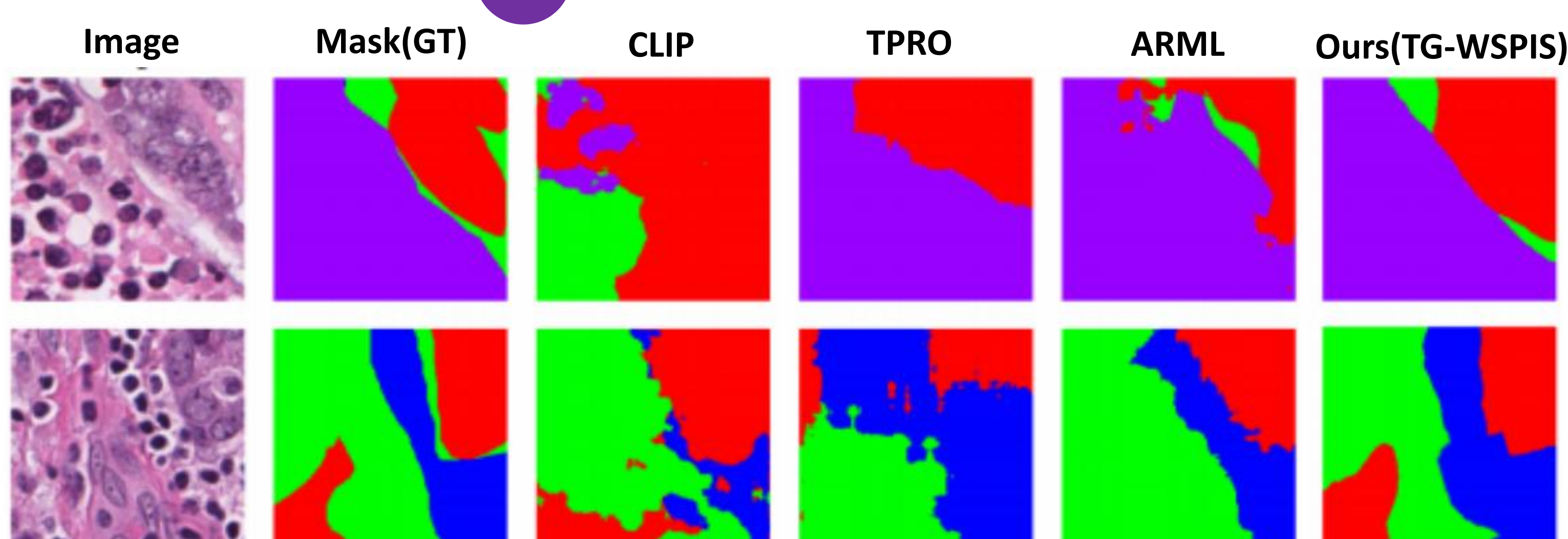
Hard + fused info + MI (baseline)
63.54 mIoU

PMCC + global proportion
74.89 mIoU

PMCC + fused information
75.65 mIoU

PMCC + fused info + MI (Ours)
76.79 mIoU

7 Qualitative Analysis



8 CONCLUSION



TG-WSPIS, a multi-granular text-guided weakly supervised pathology segmentation framework, achieves state-of-the-art performance on **BCSS-T3G** and **WSSS4LUAD-T3G** via **PMCC** alignment and **GMI** collaborative learning, and remains robust under minimal supervision.